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Q1

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FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

- The first line of input contains an integer representing the number of pages in the reference string.
- The second line contains the page reference string as a sequence of integers.
- The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Completed

```
1 #include <stdio.h>
2 int main()
3 {
4     int p,f;
5     printf("Enter number of pages in reference string: \n");
6     scanf("%d",&p);
7     int r[100];
8     printf("Enter the reference string: \n");
9     for(int i=0;i<p;i++)
10    {
11        scanf("%d",&r[i]);
12    }
13    printf("Enter number of frames: \n");
14    scanf("%d",&f);
15    int frames[10];
16    int pos=0;
17    int faults=0,hit=0;
18    int found=0;
19    for(int i=0;i<p;i++)
20    {
21        frames[i]=-2;
22    }
23    for(int i=0;i<p;i++)
24    {
```

Test Cases2Run Sample CasesRun All Cases

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Sample Input 1

7

1 3 0 3 5 6 3

3

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 6

Total Page Hits: 1

How it works

Ref String | Frames | Status

25 found=0;

26 for(int j=0;j<f;j++)

27 {

28 if(frames[j]==r[i])

29 {

30 found=1;

31 hit++;

32 }

33 }

34 if(found==0)

35 {

36 frames[pos]=r[i];

37 pos=(pos+1)%f;

38 faults++;

39 }

40 }

41 printf("\nTotal Page Faults: %d\n",faults);

42 printf("Total Page Hits: %d\n",hit);

43 return 0;}

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

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Q1

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Total Page Faults: 6
Total Page Hits: 1

How it works

Ref String	Frames	Status
1	1 - -	Fault
3	1 3 -	Fault
0	1 3 0	Fault
3	1 3 0	Hit
5	5 3 0	Fault
6	5 6 0	Fault
3	5 6 3	Fault

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

Custom Test

Success

Input:

7
1 3 0 3 5 0 3
3

Expected Output:

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1

Output:

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1

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